Project Architecture

1. Purpose of the Architecture

This document presents the architecture of the entire project from the perspective of a research workflow. The repository is designed as a reproducible scientific pipeline in which simulation, interpretation, visualization, and publication are connected but clearly separated.

2. System-Level View

2.1 Plain-language pipeline (for non-specialists)

This diagram answers: **"If I run the project once, what happens from start to finish?"**

2.2 Research pipeline: ODD -> Simulation -> LLM -> Results

This is the clearest end-to-end flow if you want to explain the study to someone outside the field:

In plain words: - `ODD` defines the model structure and the meaning of the numbers. - The simulation turns those rules into agent behavior over time. - The `LLM` does not drive the model; it explains the local decision context in short text. - The outputs are then aggregated into figures and tables for the paper.

2.2 Technical system view (modules)

3. Architectural Principles

3.1 Separation of Concerns

The repository separates:

- scientific hypotheses and scenario definitions;
- agent-level simulation logic;
- provider-specific LLM access;
- figure and report generation;
- project memory and execution tracking.

3.2 Reproducibility

Generated artifacts are derived from source files and are intentionally excluded from version control. The repository is therefore designed to publish source, not ephemeral evidence bundles.

3.3 Interpretability

The architecture preserves a strong distinction between:

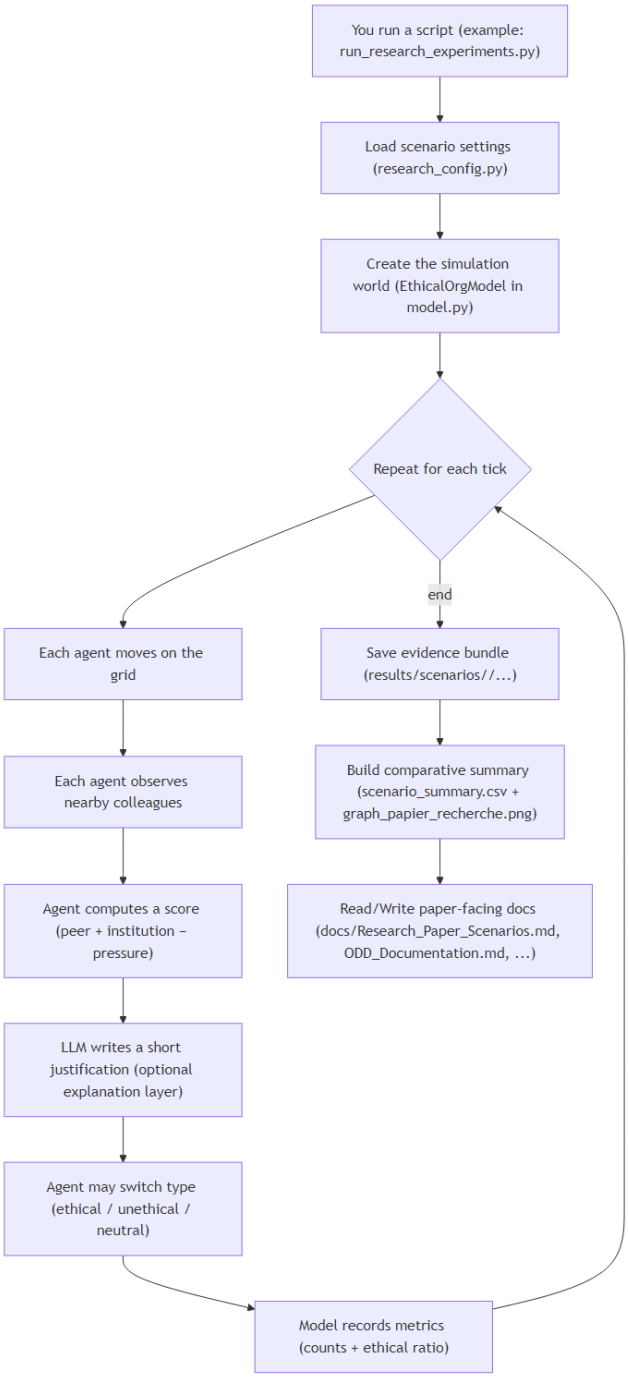

Figure 1: Mermaid diagram 1

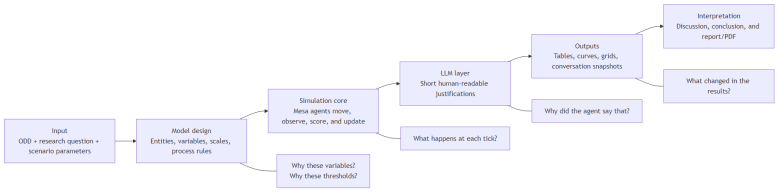

Figure 2: Mermaid diagram 2

- explicit simulation logic implemented in Python;
- interpretive textual output generated by the LLM layer.

This distinction is central to methodological transparency.

4. Runtime Architecture

4.1 Main Runtime Paths

Path	Objective	Typical entry point
Quick validation	Small-scale local run	`run_simple.py`
Standard simulation	Single scenario with charts	`run.py`
Scenario comparison	Multiple dilemmas on a shared runtime	`run_scenarios.py`
Full research campaign	Controlled structural experiments with artifact bundles	`run_research_experiments.py`
Interactive exploration	Browser-based inspection	`vis_server.py`

4.2 Runtime Data Flow

4.3 What happens inside one tick (detailed)

5. Research Architecture

The research campaign is intentionally organized around scenario bundles.

This design makes each scenario auditable as an independent evidence package.

6. Documentation Architecture

6.1 Source Documents

The `docs/` folder now serves distinct documentation audiences:

- `Execution_Guide.md` for operational onboarding;
- `Maintenance_et_Lancement.md` for rerun and maintenance procedures;
- `ODD_Documentation.md` for the formal model description;
- `Code_Architecture.md` for code-level structure and function interactions;
- `Project_Architecture.md` for the global system view.

6.2 Publication Path

7. Provider-Agnostic LLM Architecture

The repository is designed so that future users can choose the provider that matches their institutional, financial, or methodological constraints.

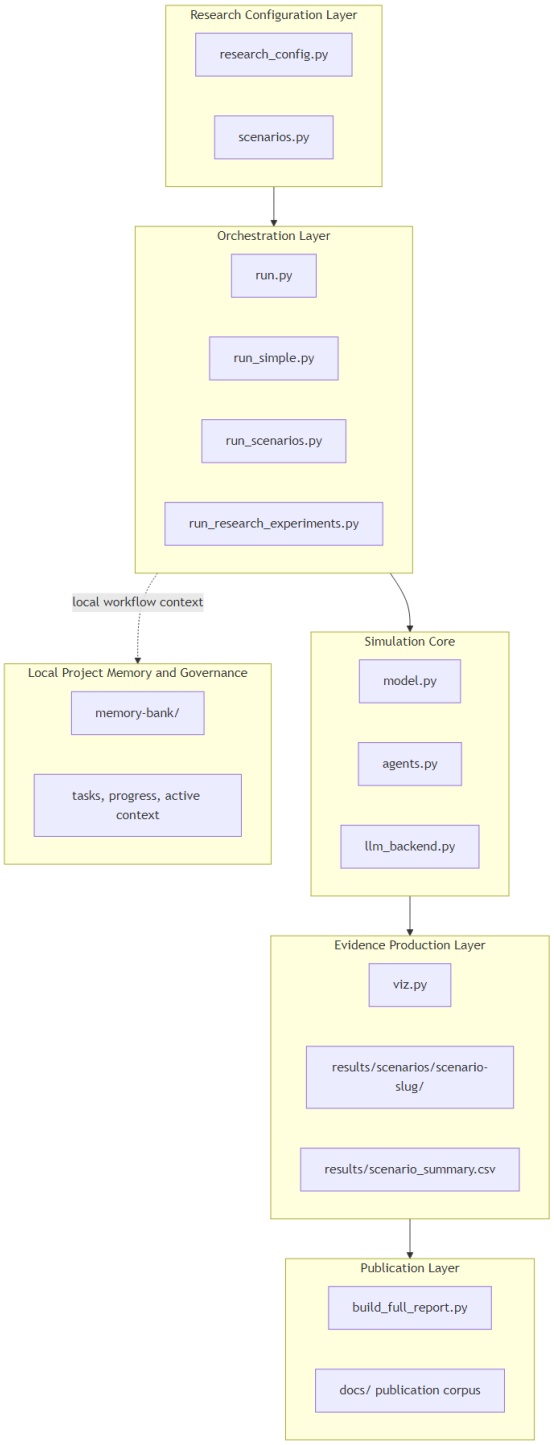

Figure 3: Mermaid diagram 3

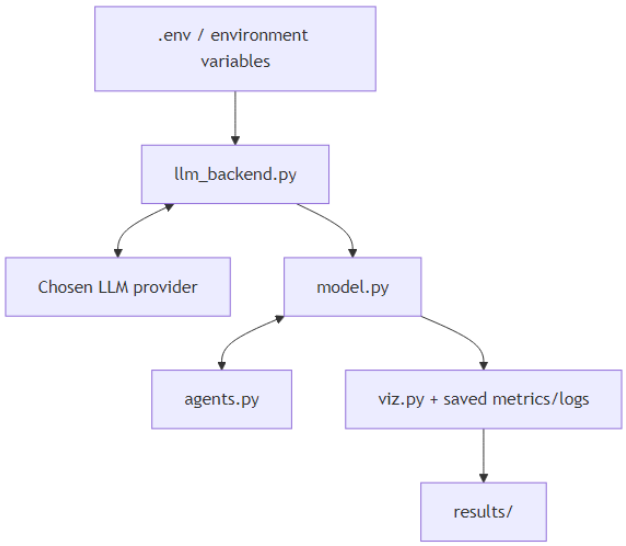

Figure 4: Mermaid diagram 4

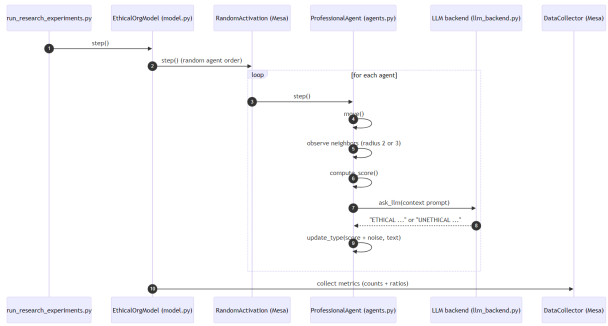

Figure 5: Mermaid diagram 5

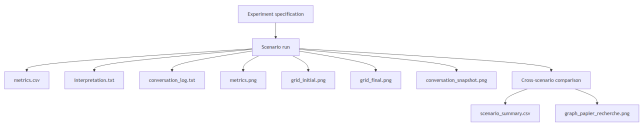

Figure 6: Mermaid diagram 6

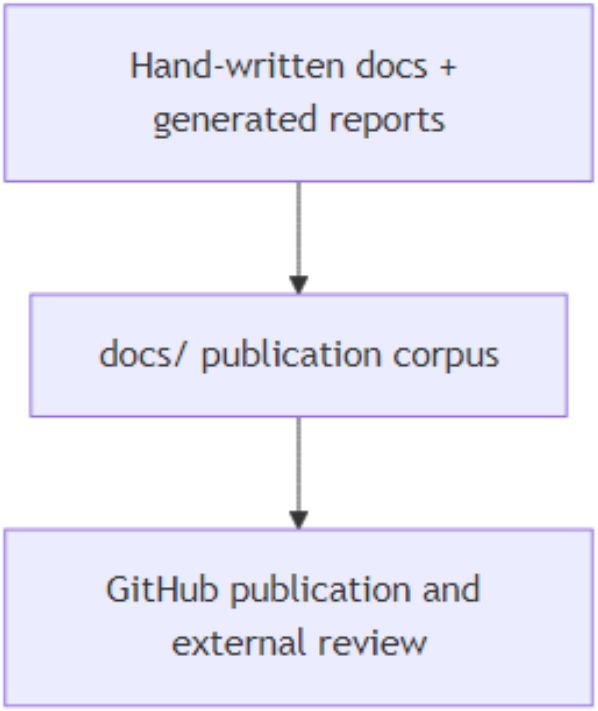

Figure 7: Mermaid diagram 7

7.1 Contract

The simulation code expects only:

- a model identifier;
- an API key;
- an optional provider label;
- an optional compatible base URL.

7.2 Encapsulation

All provider-facing logic is contained inside `llm_backend.py`. This reduces coupling and prevents the simulation core from becoming tied to a single vendor or SDK.

8. Repository Topology

```
/
|- agents.py
|- model.py
|- llm_backend.py
|- research_config.py
|- run*.py
|- viz.py
|- build_full_report.py
|- docs/
|  |- Execution_Guide.md
|  |- Maintenance_et_Lancement.md
|  |- ODD_Documentation.md
|  |- Code_Architecture.md
|  |- Project_Architecture.md
|- memory-bank/
|  |- projectbrief.md
|  |- productContext.md
|  |- activeContext.md
|  |- systemPatterns.md
|  |- techContext.md
|  |- progress.md
|  |- tasks/
|- results/              generated locally, ignored by git
```

9. Why This Architecture Supports Research Quality

This structure is appropriate for research-oriented work because it supports:

- traceable scenario definitions;
- clear separation between model assumptions and narrative outputs;
- reproducible evidence generation;

- independent review of figures, logs, and reports;
- publication-ready documentation without polluting the repository with generated files.

10. Recommended Reading Order

For a new reader or evaluator, the recommended order is:

1. `README.md`
2. `docs/Project_Architecture.md`
3. `docs/Code_Architecture.md`
4. `docs/ODD_Documentation.md`
5. `docs/Execution_Guide.md`

This sequence moves from strategic understanding to implementation detail and finally to execution practice.